

WEEK 1 · TASK 2

NETWORKING BASICS

IP • Ports • Protocols • OSI Model

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Task Week 1 · Task 2	Focus Networking Foundations	Platform TryHackMe + Windows
Work Completed Practical IP check + networking study	Level Foundation	Evidence Screenshots

1. Introduction

This report covers the basic networking concepts completed for Week 1, Task 2. It includes IP addresses, MAC addresses, common ports, TCP and UDP, the seven-layer OSI model, and a basic Windows network configuration check.

2. Learning Objectives

- Understand the purpose of IP and MAC addresses.
- Learn the role of common network ports and services.
- Understand the main differences between TCP and UDP.
- Learn the seven layers of the OSI model.
- Use ipconfig to view network information on Windows.

3. Practical Environment

Operating System: Windows

Command used: ipconfig

Learning platform: TryHackMe

4. Private IP Address — Practical Work

The ipconfig command was used to view the computer's network configuration. The screenshot shows private IPv4 addresses using the 192.168.x.x range, with a subnet mask of 255.255.255.0.

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Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::1598:2498:2efb:2b1a%3
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::92db:93dc:65a6:266f%11
    IPv4 Address. . . . . : 192.168.77.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Anslutning till lokalt nätverk* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Anslutning till lokalt nätverk* 2:
```

Figure 1. Windows IP configuration showing private IPv4 addresses and subnet masks.

5. IP Address & MAC Address

IP address: A logical address used to identify a device or network interface. IPv4 uses 32 bits, while IPv6 uses 128 bits.

MAC address: An address associated with a network interface and mainly used for communication on the local network.

Feature	IP Address	MAC Address
Purpose	Logical network identification	Local network interface identification
OSI layer	Layer 3 — Network	Layer 2 — Data Link
Example	192.168.1.10	00-1A-2B-3C-4D-5E

6. IPv4 vs IPv6

IPv4	IPv6
32-bit address	128-bit address
Example: 192.168.1.10	Example: 2001:db8::10
Smaller address space	Much larger address space

7. The 7-Layer OSI Model

The OSI model is a seven-layer framework that helps explain how data moves between networked devices.

Layer	Name	Example	Simple purpose
7	Application	HTTP / DNS	Network services used by applications.
6	Presentation	TLS / encoding	Data format, encryption and compression.
5	Session	Session management	Creates and manages communication sessions.
4	Transport	TCP / UDP	End-to-end data delivery.
3	Network	IP / routing	Moves packets between networks.
2	Data Link	Ethernet / MAC	Local delivery using frames and MAC addresses.
1	Physical	Cable / radio	Transmits raw bits.

Easy order: Application → Presentation → Session → Transport → Network → Data Link → Physical.

8. Encapsulation

When data is sent across a network, each layer adds information needed for communication. At the receiving side, this information is processed in reverse order.

Upper layers	Data
Transport	Segment / Datagram
Network	Packet
Data Link	Frame
Physical	Bits

9. Common Network Ports & Services

Port	Service	Purpose
20	FTP Data	File transfer data channel
21	FTP Control	FTP command and control channel
22	SSH	Secure remote access
23	Telnet	Remote terminal access
25	SMTP	Sending email
53	DNS	Domain name resolution
80	HTTP	Web traffic
443	HTTPS	Web traffic protected with TLS
110	POP3	Receiving email
143	IMAP	Accessing and synchronizing email

10. TCP vs UDP

TCP	UDP
Connection-oriented	Connectionless
Reliable delivery with acknowledgements and retransmission	No built-in guarantee of delivery
More overhead	Lower overhead
Examples: HTTPS, SSH, FTP	Examples: DNS, streaming, VoIP

11. TryHackMe Evidence

What is Networking? — The room was completed at 100%.

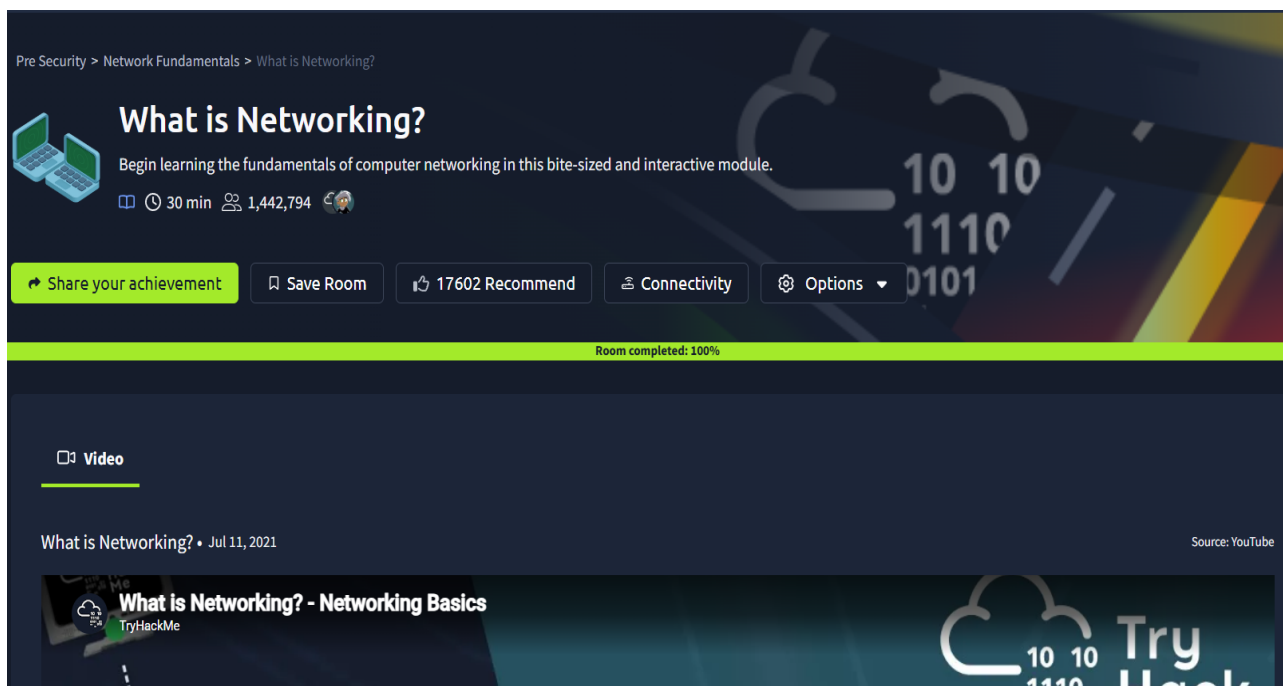


Figure 2. TryHackMe — What is Networking? — 100% completion.

Introductory Networking — The room was completed at 100%. It includes the OSI Model overview, Encapsulation, and the TCP/IP Model.

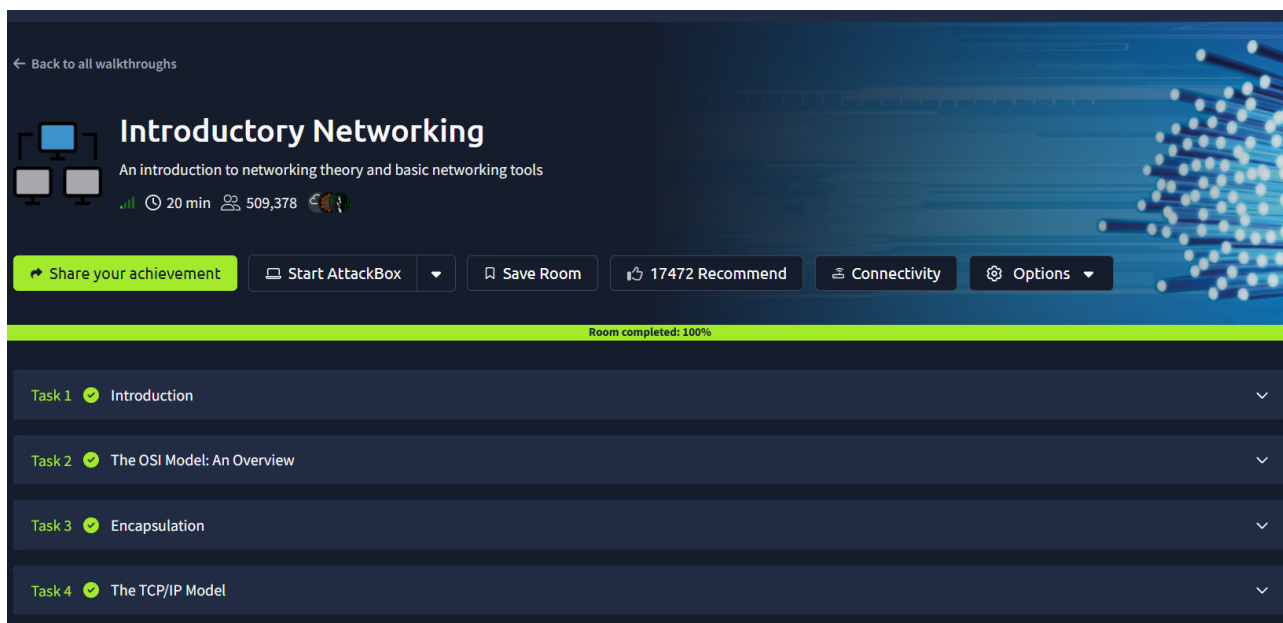


Figure 3. TryHackMe — Introductory Networking — 100% completion.

12. Key Learnings

IP addressing: Private IPv4 addresses can identify interfaces within a local network.

MAC addressing: MAC addresses are used mainly for communication on the local network.

Ports: Ports identify services such as SSH, DNS, HTTP and HTTPS.

TCP and UDP: TCP focuses on reliable delivery, while UDP has lower overhead and does not guarantee delivery.

OSI model: The seven layers describe different parts of network communication.

Encapsulation: Networking information is added as data moves down the protocol stack.

13. Conclusion

This task provided a foundation in computer networking through practical Windows work and TryHackMe learning. The main concepts covered were IP addresses, MAC addresses, ports, protocols, TCP and UDP, the OSI model, and encapsulation.